



Society of Petroleum Engineers

SPE-228899-MS

Enhanced Zonal Isolation for Water Shutoff and Gas Recovery in Depleted Sandstone Reservoirs, A Case Study

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Excessive water production in depleted gas reservoirs, especially within multilayer systems exhibiting crossflow, poses significant challenges to optimal gas recovery. This study details a targeted zonal isolation strategy in two wells, employing acid-soluble cement slurry and a specially formulated calcite-based plug reinforced with a reservoir-friendly, non-damaging gel. This approach effectively isolated water-producing perforations while preserving gas-bearing zones, overcoming challenges such as narrow inter-perforation gaps and wellbore losses, thereby enhancing gas output.

Initial production from first well showed strong gas deliverability. Continued drawdown led to reservoir depletion, water breakthrough, and rising water cut. PSP diagnostics identified dominant gas from upper perforations, water influx from the third set during flow, and gas crossflow from the first to second perforations during shut in. To isolate the water-bearing zone, a gas-tight, acid-soluble cement plug was accurately placed via coiled tubing below the gas zone, using tailored lost circulation materials and optimized pumping parameters to reduce wellbore losses. In the second well, staged zonal isolation with a calcite plug via coiled tubing effectively sealed the water influx.

Cement slurry for the first well was precisely engineered as a high-performance, acid-soluble, gas-tight system, incorporating latex for gas migration control, retarders for placement time optimization, defoamers to eliminate entrained air and ensure consistent slurry density, and fluid loss additives to minimize filtrate invasion into permeable formations and maintain slurry integrity under differential pressure during plug placement. Severe fluid losses in both wells, driven by reservoir depletion, were proactively addressed through the placement of a tailored LCM plug across the perforations, successfully restoring static wellbore conditions and enabling precise plug placement within the narrow target interval. Limited vertical separation between gas and water influx zones further demanded exact depth correlation and real-time monitoring during placement. In the first well, precise placement of the cement barrier through coiled tubing enabled successful hydraulic isolation of the water-bearing zone, validated by post-intervention diagnostics showing a 98% reduction in water production and a fourfold increase in gas recovery. In the second well, a staged approach using a calcite-based isolation plug via coiled tubing and cement slurry via wireline dump bailer effectively sealed the lower water-bearing interval, enabling individual testing of the upper gas-bearing zone

The deployment of precisely engineered gas-tight slurry systems via coiled tubing, coupled with controlled placement and targeted diagnostics, enabled a cost-effective intervention for isolating water-producing zones in depleted gas wells. This case underscores the value of designing zonal isolation treatments around reservoir-specific conditions, presenting a scalable methodology for production enhancement in mature gas fields

Introduction

In depleted gas reservoirs especially, those completed across multilayer sandstone formations excessive water production emerges as one of the most persistent and production-limiting challenges. These formations are often characterized by significant heterogeneity, comprising multiple perforated intervals across gross pay zones that may exceed 50 meters in vertical thickness. While such completions offer initial benefits by maximizing early gas drainage, they frequently become problematic over time as water ingress from select intervals often due to reservoir depletion and shifting pressure gradients begins to dominate production. This imbalance leads to rising water cuts, reduced hydrocarbon output, increased lifting costs, and the eventual impairment of well economics. In many fields across South Asia and the Middle East, over 50% of the fluids produced from mature gas wells are water, with diagnostic logging revealing that select intervals often less than half the gross perforated length are responsible for more than 70% of total water influx (Huang et al., 2018).

The dynamic nature of production from multilayered systems means that the challenge of water ingress is often not uniform. Certain zones may continue to produce dry gas, while others typically the deeper or more permeable intervals serve as the source of water. Production logging tools (PLTs) and spinner surveys play a pivotal role in diagnosing this behaviour, enabling the identification of dominant water-producing intervals and their segregation from gas-bearing zones. In such cases, the objective is not merely to block the entire production interval but to implement a **selective zonal isolation strategy** that shuts off water influx while preserving gas deliverability.

Traditional mechanical methods for zonal isolation such as bridge plugs, inflatable packers, or cement squeezes though widely used, are not always ideal. Bridge plugs and packers require precise placement and often necessitate wellbore conditions that are difficult to guarantee in aged or deviated completions. Moreover, conventional cement systems, once placed, are permanent and non-reversible, which may not be desirable in wells with active upper gas-producing intervals. Additionally, these techniques can introduce formation damage or impair future re-entry options, limiting long-term reservoir management flexibility (Pang et al., 2018; Wang et al., 2019).

Chemical-based water shutoff (WSO) solutions, on the other hand, have gained popularity due to their operational versatility, selective placement capabilities, and minimal footprint. Among these, **engineered cement systems with acid-soluble or re-enterable properties** offer a hybrid solution combining the mechanical integrity of cement with the reversibility of chemical treatments. Such systems can be placed with coiled tubing for precise depth control and are designed to dissolve in acid if future intervention is required, reducing the risk of permanent reservoir access loss (Chen et al., 2020). The cement can also be tailored with additives for gas migration control, fluid loss reduction, and slurry stability to address well-specific challenges.

Another important consideration is the challenge posed by wellbore fluid losses in depleted reservoirs. Severe underbalance conditions and depleted formation pressures increase the risk of cement slurry losses during placement, undermining the effectiveness of the isolation job. To address this, pre-treatment with LCM pills, dual-string displacement techniques, and the use of gel spacers have been incorporated into field execution plans. These strategies help restore static wellbore conditions and improve plug integrity by preventing slurry channelling or loss into the formation during setting. Laboratory testing also plays a key role in designing the cement formulation optimizing parameters such as thickening time, compressive

strength development, slurry density, and fluid loss control to ensure robust performance against differential pressures under downhole conditions.

In regions such as Pakistan, Oman, and parts of India, the deployment of such advanced cement-based zonal isolation solutions has transitioned from a remedial method to a proactive reservoir management tool. The success of these interventions has also prompted operators to incorporate selective plug placement as part of routine well surveillance and re-completion planning, especially in fields where zonal production profiles vary significantly over time. The economic impact is also notable: by restoring gas production and minimizing produced water handling costs, these treatments offer significant ROI in marginal fields and extend the productive life of the assets.

This paper presents a detailed case study of two different wells from such mature sandstone gas fields located in Lower Indus Basin in Southern Region of Pakistan, where a selective acid-soluble cement plug was deployed to isolate high water-producing intervals identified through PLT diagnostics. The intervention strategy was based on precise zonal targeting, customized slurry design, and mitigation of severe wellbore losses. The outcome included a significant increase in gas production and near-complete elimination of water influx, validating the effectiveness of this approach under field conditions.

Case-1

A mature gas well, completed across a thick multilayered sandstone interval, began to experience a progressive decline in hydrocarbon productivity due to an escalating water cut. Initial production performance demonstrated reliable gas deliverability, with rates exceeding 10 MMSCFD at a wellhead pressure of 1690 psi. However, over time, the well exhibited a sharp increase in water production reaching up to 1500 BPD which significantly eroded gas recovery and impacted the field's economic viability.

To understand the source of the imbalance, Production Logging Tool survey, complemented by temperature profiling, was carried out. The diagnostics revealed a clear separation in flow behaviour: the uppermost perforated interval was found to be the primary gas contributor, accounting for over 89% of total production, while the lower sets were largely responsible for water influx contributing more than 70% of the produced water. This strong disparity between gas and water-producing zones made the well an ideal candidate for targeted water shut-off.

Several zonal isolation techniques were evaluated, including calcite-based plugging, conventional cement plugs, inflatable packers, and acid-soluble cement formulations. After technical screening and risk assessment, the operator opted for an acid-soluble cement system. This choice was driven by its unique formulation properties such as controlled thickening time, high compressive strength development, and the ability to dissolve in post-job acid washes offering an effective means to isolate the offending water-bearing interval while minimizing risk to the gas-producing zone. The system also preserved optionality for future interventions by allowing controlled access through the cemented interval if required.

Several engineering considerations influenced the job design. These included anticipated loss zones, the need for controlled slurry placement in scenario of scaling, and the potential for gas migration during or after placement. The cement system was tailored to exhibit right-angle set behaviour under elevated bottomhole conditions, ensuring rapid transition from slurry to set state to mitigate the risk of gas percolation. Spacer systems were also specially designed to condition the wellbore and manage interfacial compatibility.

The job was planned for execution using coiled tubing to provide accurate depth placement within the wellbore targeting the water producing interval. The overall strategy was built around leveraging engineered fluid systems, proactive loss management, and zonal selectivity to restore gas production while eliminating excessive water cut.

Operational Challenges and Mitigation Measures

One of the most critical challenges encountered during the operation was the uncontrolled gas influx while attempting to kill the well. The influx required on-the-fly pressure management using continuous brine pumping and chemical inhibition to prevent string corrosion. Another major obstacle was the significant wellbore fluid loss encountered during displacement operations, which posed a risk of premature cement dehydration, slurry fallback, and channelling that could compromise plug integrity. In the event of uncontrolled losses, there was also a high possibility that the cement plug might fail to cover the entire perforated interval, leading to ineffective isolation of the water-producing zone. This was mitigated through the pre-placement of LCM pills, brine displacement from both CT and annulus, and strategic soaking time to restore near-static conditions.

To ensure proper cement placement over the irregular and porous surface of the settled LCM, a tailored spacer system was designed and strategically pumped both ahead of and behind the cement slurry. The lead spacer acted to condition the wellbore, bridging micro-channels, minimizing fluid loss, and stabilizing solids to prevent slurry fallback during placement. The tail spacer served as a buffer to prevent slurry contamination from the brine, preserving cement integrity during displacement. The spacer formulation included carefully selected additives to enhance particle suspension, control fluid migration, eliminate entrained air, and maintain favourable rheological properties throughout the job.

Additionally, downhole gas migration posed a serious threat to the integrity of the cement plug. To counter this, a dual-displacement sequence was employed, and annular pressures were carefully controlled to squeeze formation gas back into the reservoir prior to cement placement. The cement slurry was specifically engineered with right-angle set properties, ensuring rapid gelation after placement to minimize gas invasion during the critical transition phase. This design characteristic not only enhanced the plug's sealing capability but also reduced the risk of micro-annuli and gas tracking. Furthermore, the plug's acid solubility added a layer of operational flexibility, offering a reversible option in case of inadvertent zonal damage or future interventions.

The execution of this intervention not only restored the gas deliverability of the well but also validated the field application of engineered cement systems in complex, multi-perforated sandstone reservoirs. The post-job production surge and near-total water cut elimination underscored the value of reservoir diagnostics, adaptive engineering, and selective isolation in optimizing mature gas assets.

Job Execution

CT Run (Well Killing, Loss cure and Cement Plug Placement)

The operation commenced with the deployment of coiled tubing into the well, the operation commenced with the deployment of coiled tubing into the well, with continuous trickling of 4% NH_4Cl brine at 0.4 bpm throughout the run-in-hole. The CT was gradually run in hole until tagging PBTD at 11,695 ft. At this point, the CT was positioned just above the top of the water-producing perforations to initiate the well-killing operation. A kill fluid 4% NH_4Cl brine was pumped through CT at 1.1 bpm to establish brine returns and begin displacing gas from the wellbore. This brine, with a density of 8.41 ppg, was specifically selected to provide an overbalance of approximately 1100 psi over the estimated formation pressure, ensuring effective well control while minimizing the risk of gas percolation that could ultimately contaminate the cement plug. Once returns were observed and CT was pulled up to 11,180 ft, surface gas influx was detected, prompting immediate pumping of a 5 bbl corrosion inhibitor batch to protect the CT string from souring and gas-induced damage.

Once brine circulation was established, the well was shut in and monitored for 2 hours to observe any rise in wellhead shut-in pressure (WHSIP). No pressure build-up was detected, suggesting that gas influx was under control. However, upon reopening the well and resuming pumping, no fluid returns were observed

at surface indicating severe formation losses and confirming the need for remedial action prior to cement placement., a 15 bbl Lost Circulation Material pill was prepared and bull-headed through the CT-tubing annulus at a rate of 1.8 bpm. During the bull heading operation, the CT was continuously reciprocated to monitor string weights and ensure free movement, as any potential bridging of LCM within the CT-tubing annulus could result in coiled tubing becoming stuck. A follow-up displacement with 4% NH_4Cl brine was conducted simultaneously from CT-tubing annulus to ensure effective placement and pack-off of the LCM in the loss zones. A noticeable increase in wellhead pressure to 2000 psi and alignment of pumped volume with calculated displacement confirmed successful LCM placement. CT was then held stationary at 10,000 ft for a two-hour soaking period, with the well shut in to allow time for the LCM to stabilize in the formation.

Subsequently, CT was run in hole again to tag the top of the LCM plug, which was encountered at 11,368 ft. This depth was reconfirmed to validate LCM position before opening the well to check for losses. Returns were established after pumping 2 bbls of brine, indicating complete fluid containment within the well. While the initial plan was to clean out the LCM down to 11,668 ft, the CT could not penetrate beyond 11,387 ft due to settling and hardening of the LCM. It was therefore decided to utilize the tagged LCM top at 11,387 ft as a mechanical base for the cement plug placement. The objective was to set the top of cement at 11,267 ft to ensure full coverage of the water-bearing perforations.

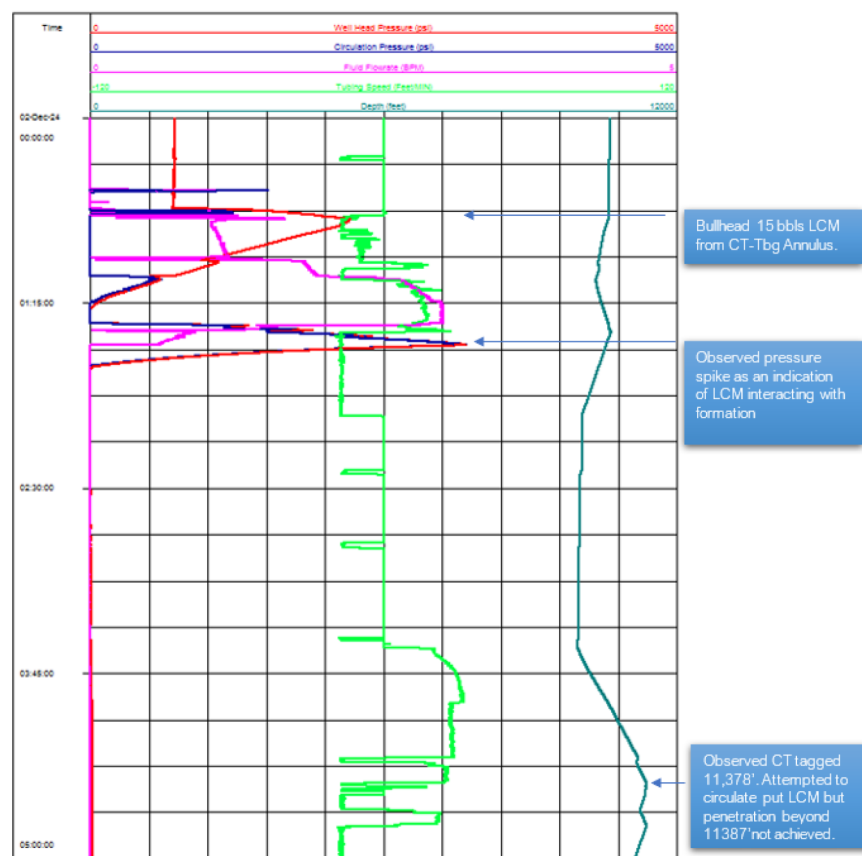


Figure 1—LCM bull heading to cure the losses

Preparation for cement placement began with the mixing of a well shielding LCM spacer. At this stage, a brief gas influx was noted, as evidenced by brine flare at the surface pits. In response, brine pumping was resumed from both the CT and annular side to stabilize well conditions, and formation gas was squeezed back into the reservoir. Once the well was confirmed static and under control, cement operations proceeded.

The acid-soluble cement slurry was prepared in accordance with the lab-qualified design. CT was positioned at 11,387 ft, where 8 bbls of engineered spacer was pumped ahead of the cement to condition

the wellbore and promote optimal slurry placement across the irregular LCM surface. This was followed by 4.5 bbls of cement slurry 1 bbl was nozzled out at 11,387 ft, while the remaining volume was pumped as CT advanced to the final placement depth at 11,267 ft. 3 bbls of spacer was pumped and kept behind the cement to prevent contamination from displacement brine and to clean the CT. Finally, the CT was picked up to 11,260 ft to perform a complete bottom-up circulation and ensure the removal of any excess cement slurry from the tubing–casing annulus.

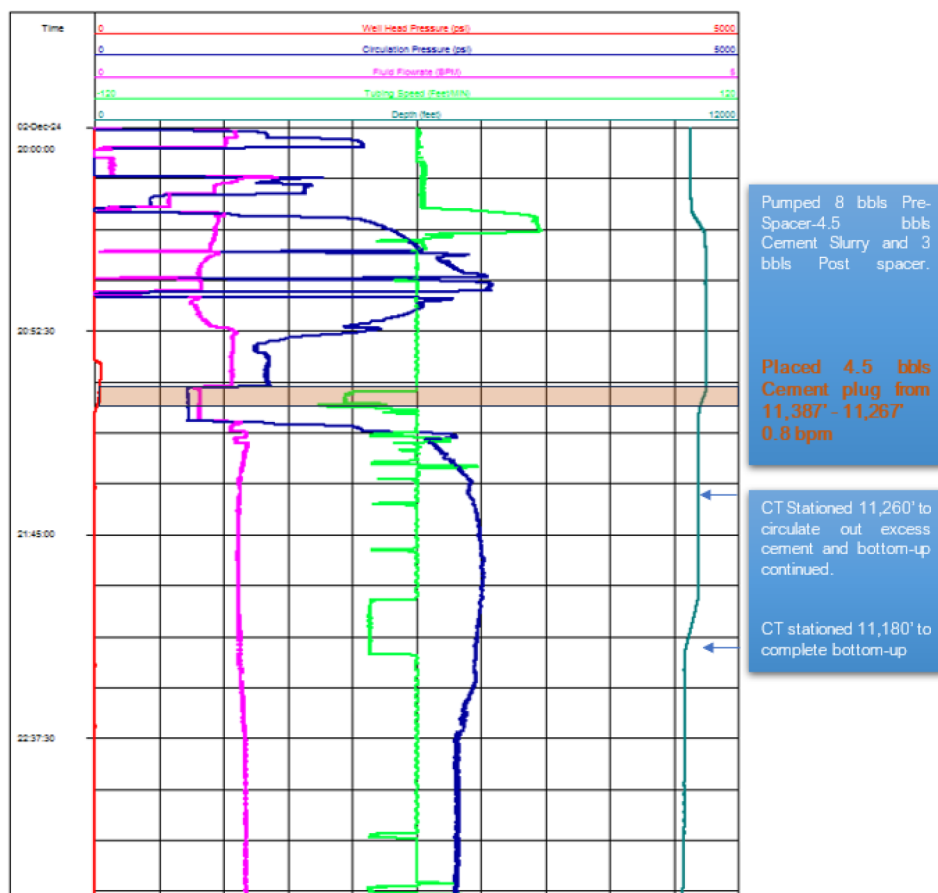


Figure 2—Acid Soluble Cement Placement across water producing interval

24-hour wait-on-cement (WOC) period was observed to ensure the development of adequate compressive strength. Following the WOC period, a slickline run confirmed the top of cement at 11,267 ft, verifying accurate placement and plug integrity. To reinitiate flow and clean up the wellbore, a nitrogen kick-off was conducted via coiled tubing from 3,500 ft. A clear gas influx was observed at surface, indicating the successful restoration of communication with the upper gas-producing interval. Subsequently, the coiled tubing was pulled out of hole, and the operation was concluded.

Post-job flowback results demonstrated a substantial improvement in well performance. Gas production surged from 2.2 to 8.4 MMSCFD, while water cut dropped drastically from 3000 BPD to just 34 BPD confirming the effective isolation of the water-producing intervals and the reestablishment of efficient gas deliverability.

Case-2

This case involves the successful zonal isolation of a lower perforated interval that was previously contributing to unwanted water influx. The objective was to hydraulically disconnect this interval to enable future evaluation of overlying zones without interference. A dual-barrier approach was selected, combining

an acid-soluble calcite slurry with a wireline-placed cement plug via dump bailer, to ensure both operational flexibility and long-term isolation integrity.

The isolation design included the deployment of a 40-barrel calcium carbonate slurry placed in two stages through coiled tubing. This system was chosen due to its compatibility with potential future re-entry operations, given that calcium carbonate is dissolvable in acid. Placement operations were adapted to field constraints, including limited water availability and tubing ID restrictions. The coiled tubing was carefully run to the targeted depth, and the slurry was pumped at controlled rates with adequate settling time between each stage. After placement, the top of the calcite barrier was tagged with verified slackoff, indicating proper placement and plug stability at the desired depth.

Following the calcite slurry placement, a secondary barrier was introduced via a wireline cement dump bailer. The cement system was delivered in five incremental runs, with wait-on-cement periods of 15 hours after the first three runs and 36 hours after the final two. This staged cementing technique allowed for proper hydration and setting under static conditions. The tagged top of cement was located within acceptable proximity to the planned depth, confirming vertical coverage of the previously watered out perforated interval.

Confirmation of isolation integrity came from slickline-deployed memory gauge data and pressure build-up analysis. These tools recorded pressure behaviour over extended shut-in periods and provided a clear indication that no pressure transfer or fluid migration was occurring from the sealed interval. The pressure profile remained stable and isolated, affirming the barrier's effectiveness.

Importantly, there were no signs of upward fluid movement, crossflow, or pressure imbalance during the entire post-placement monitoring phase. All evidence pointed to successful hydraulic disconnection of the lower zone, with both the calcite and cement barriers holding as intended.

This case demonstrated that combining an acid-soluble particulate slurry with wireline dump cementing can deliver reliable zonal isolation in mature gas wells. The use of a calcite plug offered the benefit of future removability without introducing significant formation damage, while the cement plug provided structural assurance and redundancy. Together, the two barriers ensured a high degree of confidence in isolating the unwanted water producing interval.

In conclusion, the lower perforated interval was successfully and permanently isolated using a two-part strategy, with no indications of communication or barrier failure post-execution. The design and diagnostics applied in this case offer a reliable reference for zonal isolation in similar conditions, particularly where selective reservoir evaluation or water shutoff is required.

Conclusions

- The application of **acid-soluble cement systems** introduced a transformational approach to zonal isolation by enabling **selective shutoff of water-producing zones** while preserving gas deliverability. Unlike conventional plugs, this system offered **reversible isolation**, minimizing permanent damage to gas-bearing intervals and allowing future re-entry if needed.
- The integration of **diagnostic PLT data with responsive cement system design** allowed for targeted intervention, reducing unnecessary formation exposure and preserving long-term reservoir value, highlighting the role of zonal isolation in **sustainable field development**.
- Using **lab-tested, high-performance cement blends** with tailored thickening time, compressive strength development, and gas migration control additives, the job execution remained resilient under severe loss circulation and high gas influx conditions.
- The **plug design philosophy shifted from reactive to proactive**, establishing cement plugs as a reservoir management tool instead of a last-resort remedial option.
- Post-job results demonstrated a **major uplift in well performance**, with gas production surging from 2.2 to 8.4 MMSCFD and water cut reduced from 3,000 to 34 BPD, validating the effectiveness

of the intervention and its **tangible economic benefits** in terms of gas recovery and water handling reduction.

- The outcome validates acid-soluble cement plugs as a scalable, field-ready strategy for tackling excess water production in multilayered gas reservoirs, with potential for wider implementation in similar mature assets across the region.

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